Osteopathic Manipulative Medicine: Osteopathic Considerations in the Hospitalized Patient

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Make.
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Session Objectives

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1. The student will understand the role of inflammation and nociception in viscerosomatic reflexes

2. The student will be able to describe the role of inflammation in the lymphatic system

3. The student will be able to identify the anatomy and physiology of the lymphatic system, including thoracic duct

4. The student will be able to identify clinical findings in acute viscerosomatic reflexes

5. The student will be able to describe how and why it is important to treat segmental facilitation from viscerosomatic reflexes



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No two patients with the same disease present the same way. No patient with the same disease presents the same way on different visits.
A.L. Wales, DO

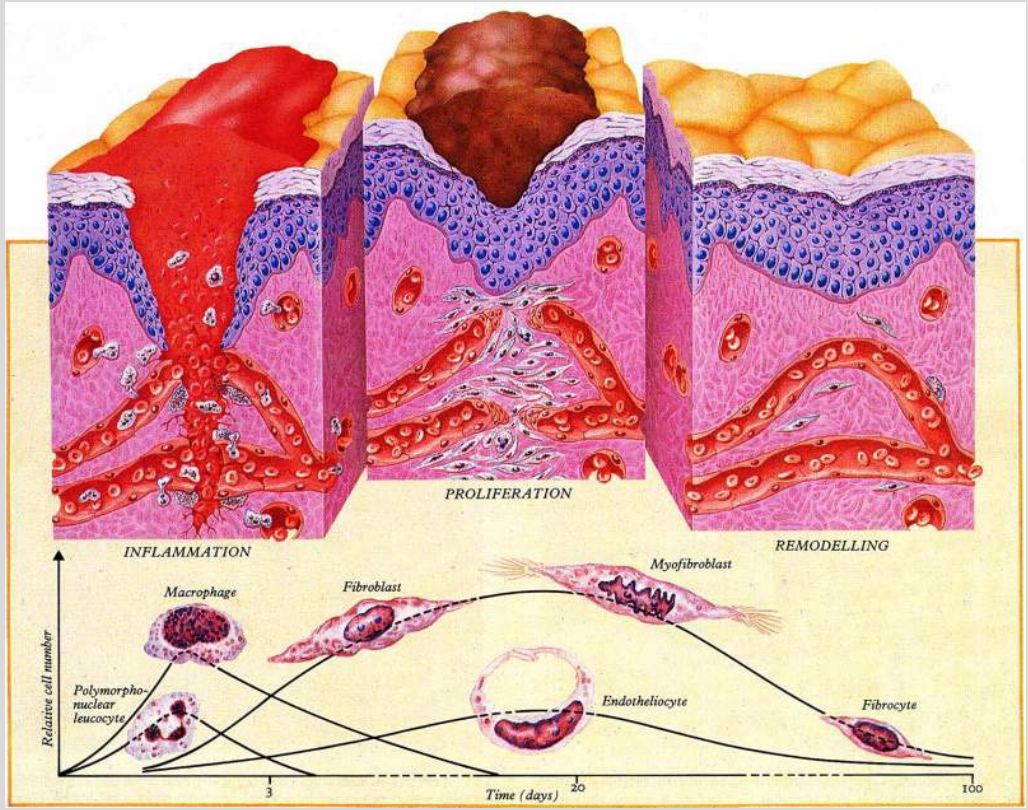
The body's response to disease

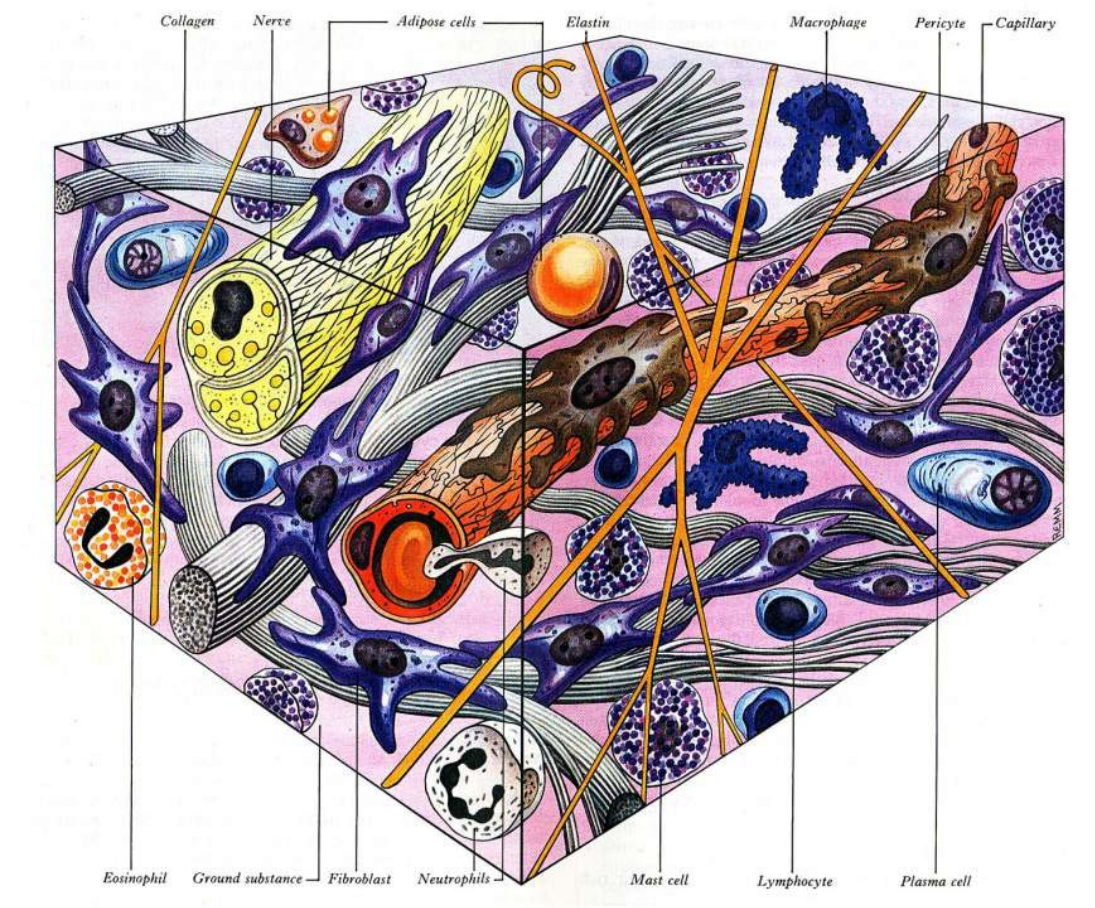
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- Local – Inflammation
- Humeral – Immune
- Nervous System – Spinal cord reflexes
- Central – Hypothalamic/pituitary/adrenal axis

Inflammation

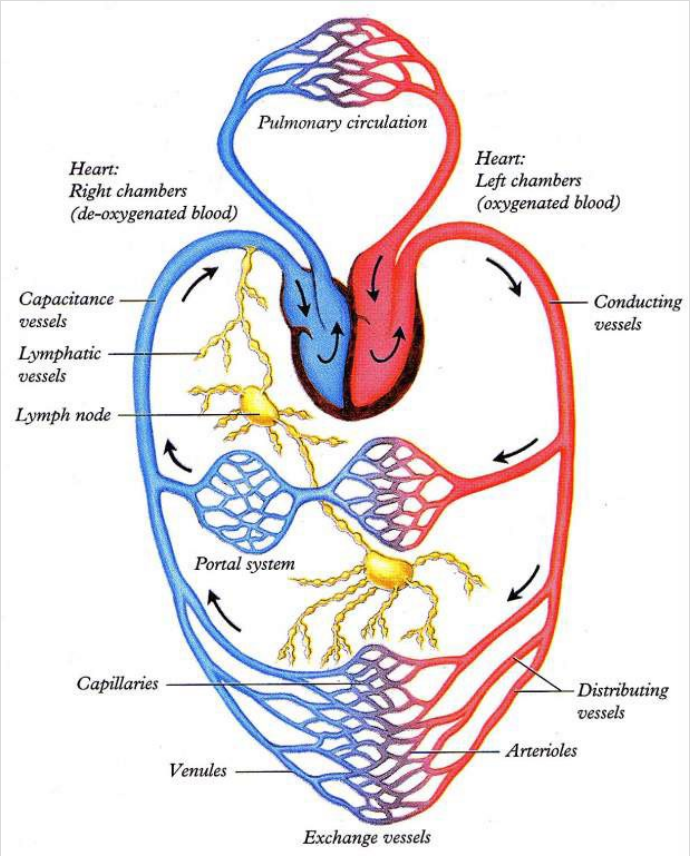
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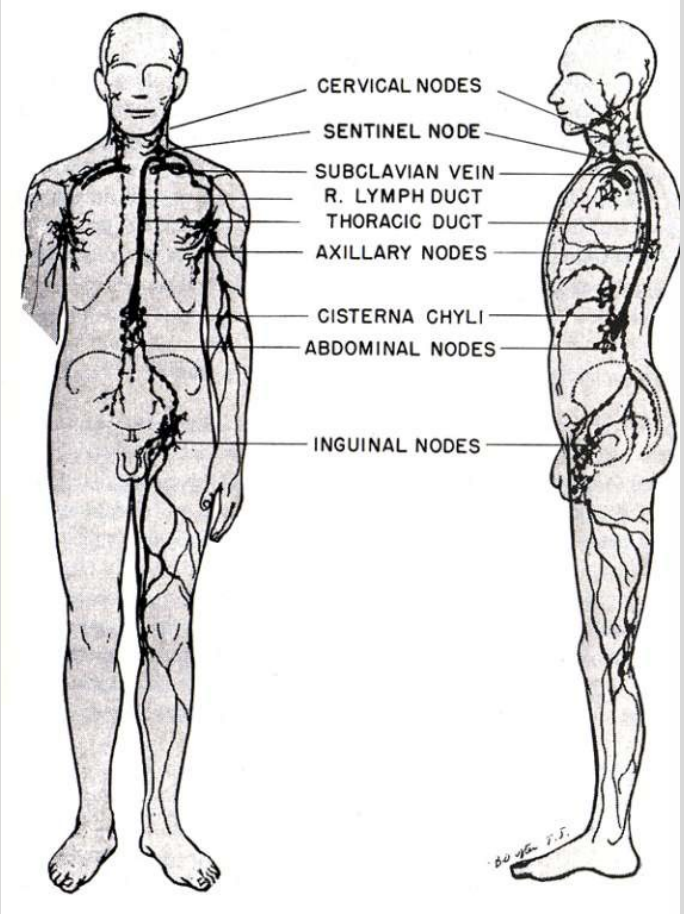

Collagen
Nerve
Adipose cells
Elastin
Macrophage
Pericyte
Capillary
Eosinophil
Ground substance
Fibroblast
Neutrophils
Mast cell
Lymphocyte
Plasma cell

Circulation

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- The Lymphatics, although often considered part of the circulatory system, is actually open to the interstitium, and functions as part of it

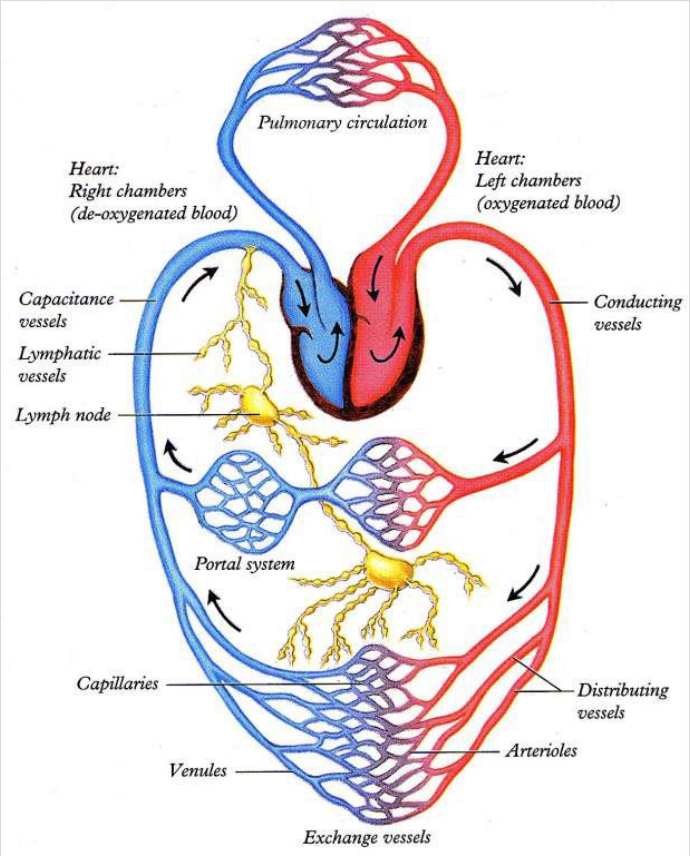


Role of lymphatics in inflammation

- Antigen – the lymphatics carry antigen to the lymph nodes where the immune response is initiated. This is particularly important in infections.

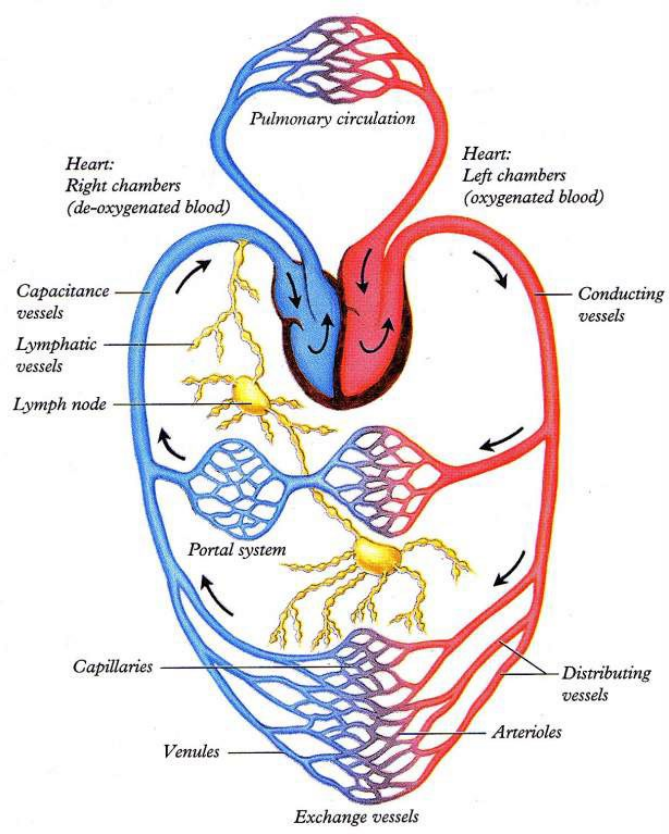
Circulation

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- Inflammation produces a change in capillary permeability which allows a large protein efflux into the interstitium.
- This produces an inflammatory exudate, which can only be drained via the lymphatics.

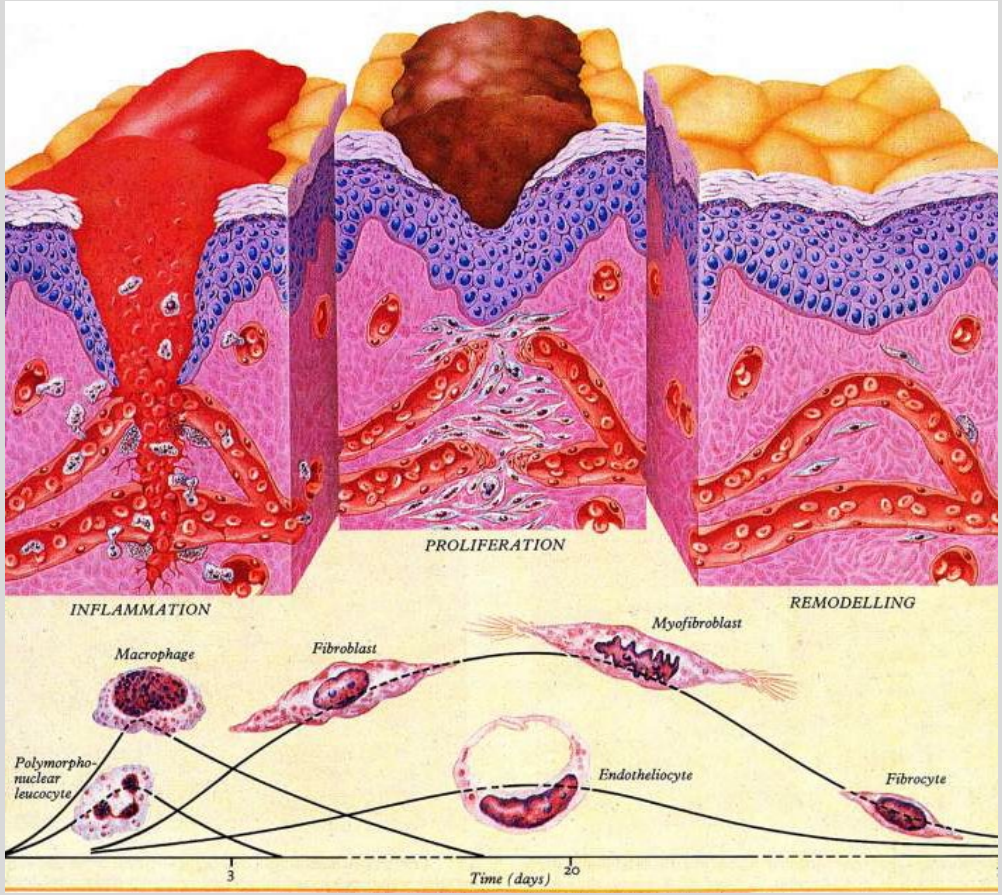
Circulation



- This also produces stasis of the local capillary circulation, so that all circulation from an inflammed tissue is lymphatic.
- Once the tissue becomes saturated with fluid, circulation to the inflammed tissue will occur at the same rate as lymphatic drainage

Resolution of the Inflammatory process

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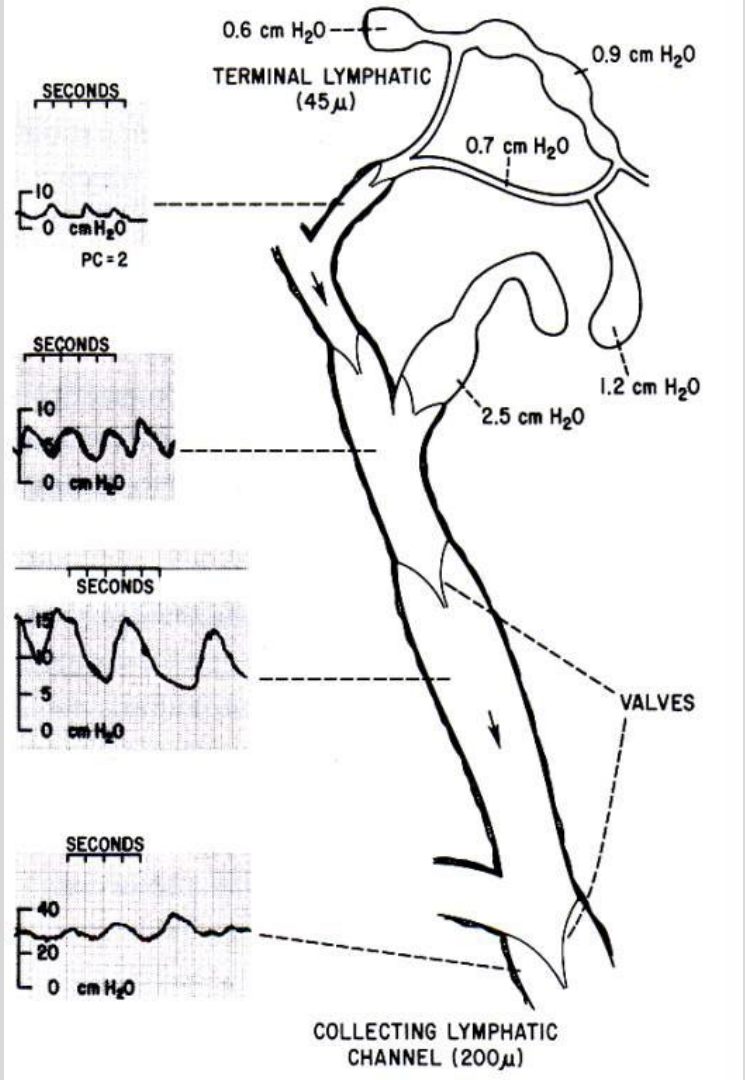


Mechanics of the lymph circulation

- Terminal – drainage of thoracic duct into the subclavian vein.
- Vascular – the movement of lymph through the lymphatic vessels.
- Formation – the movement of extracellular fluid into the initial lymphatic

Terminal lymph drainage

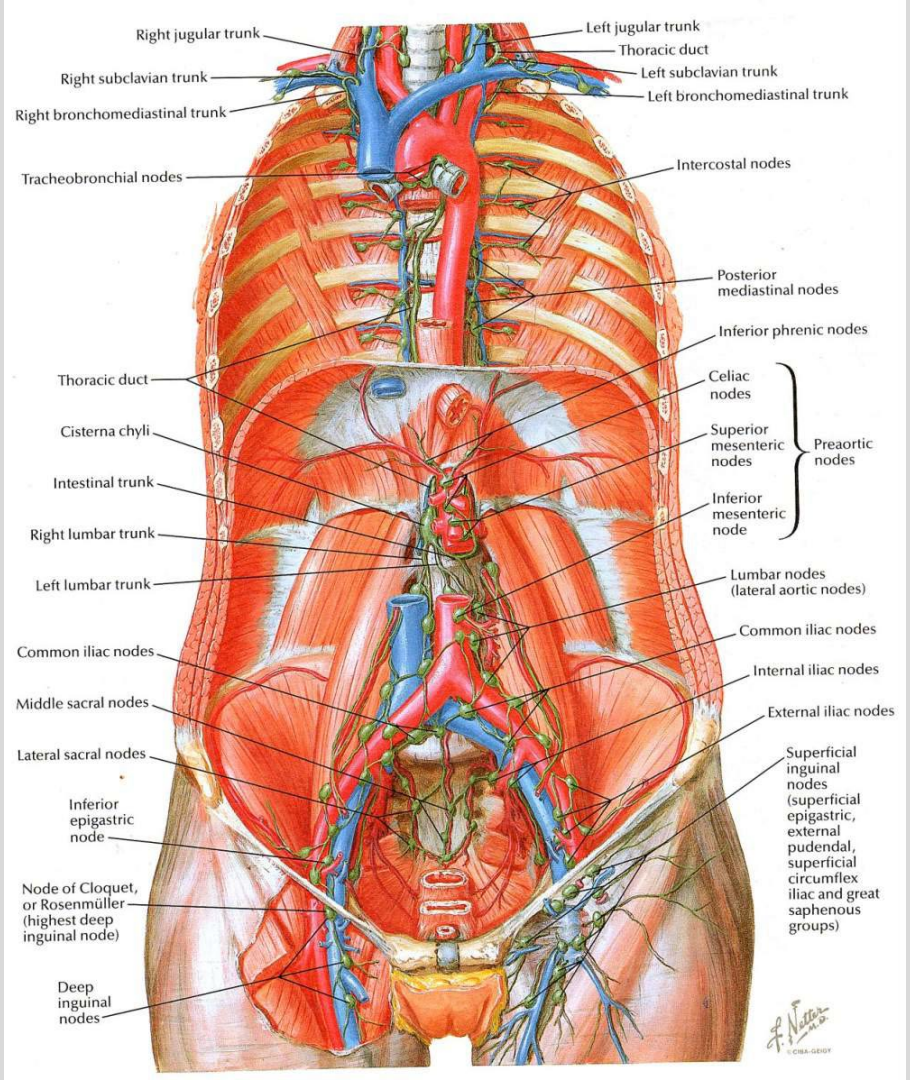
- Respiratory/circulatory function
  - J. Gordon Zink
  - 35 – 60% of thoracic duct flow is produced by respiration
    - Relation of crus to cysterna chyle
    - Venturi effect
    - Effects of respiration on lymph entering thoracic duct



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Vascular lymph flow

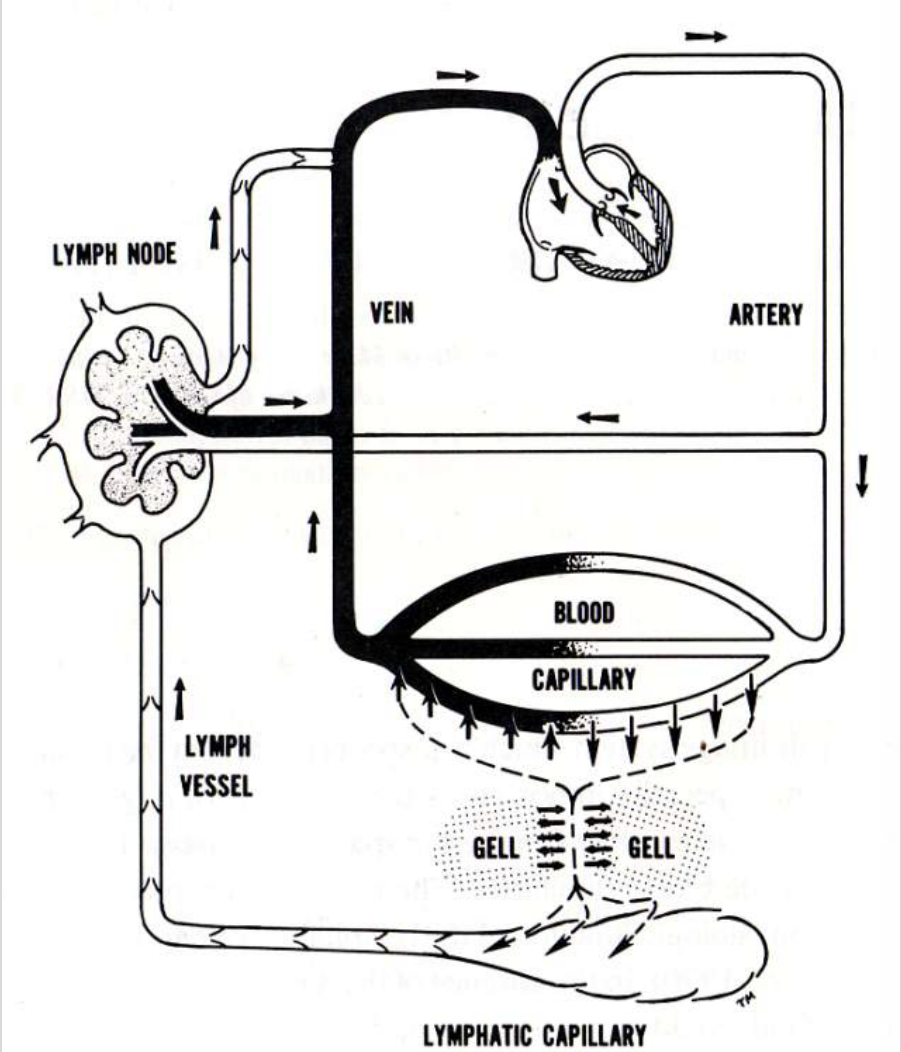
- Lymphangion – a unit with smooth muscle and valves
- Has intrinsic peristaltic action
- Can generate enough pressure to account for all intravascular lymph movement
- Can be obstructed with increased tissue tension/somatic dysfunction



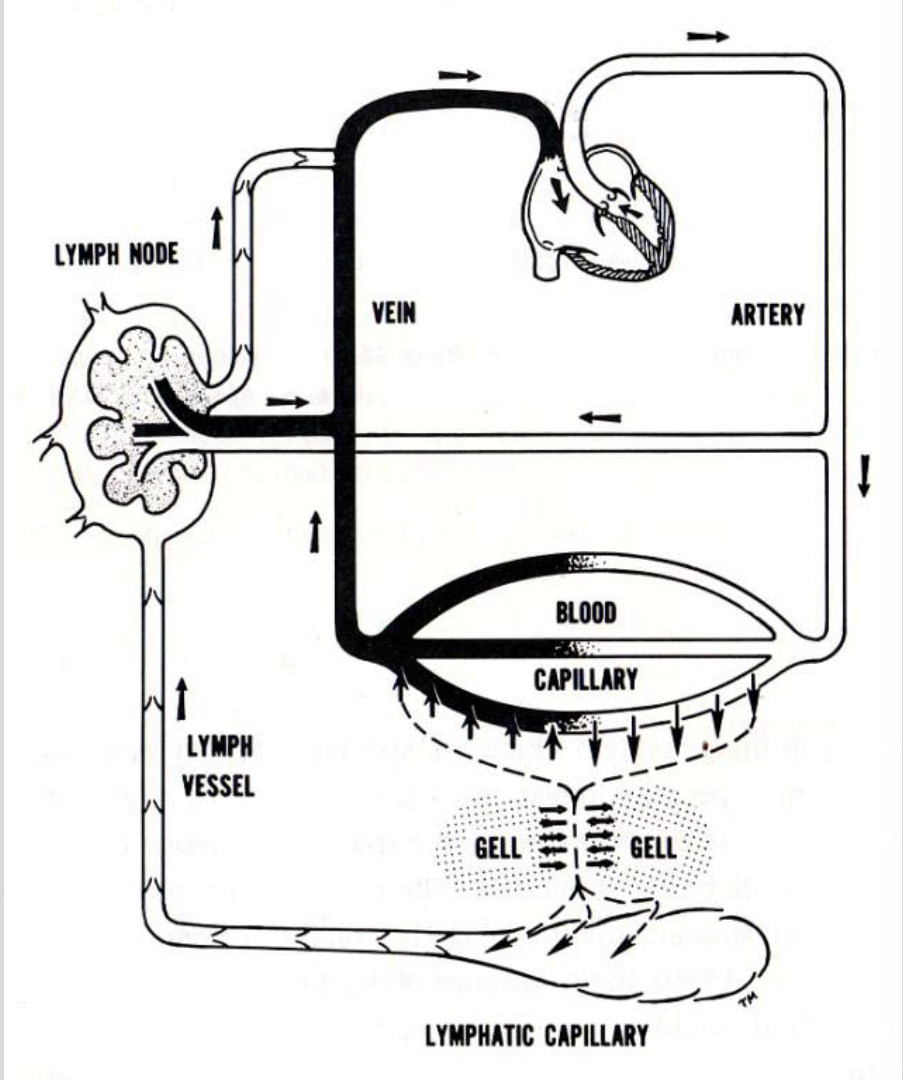
Can be obstructed with increased tissue tension/somatic dysfunction

Lymph Formation

- There is a small uphill hydrostatic gradient which must be overcome for extracellular fluid to move into an initial lymphatic
- This may be the limiting factor in overall lymph flow
- External forces are necessary for lymph formation.



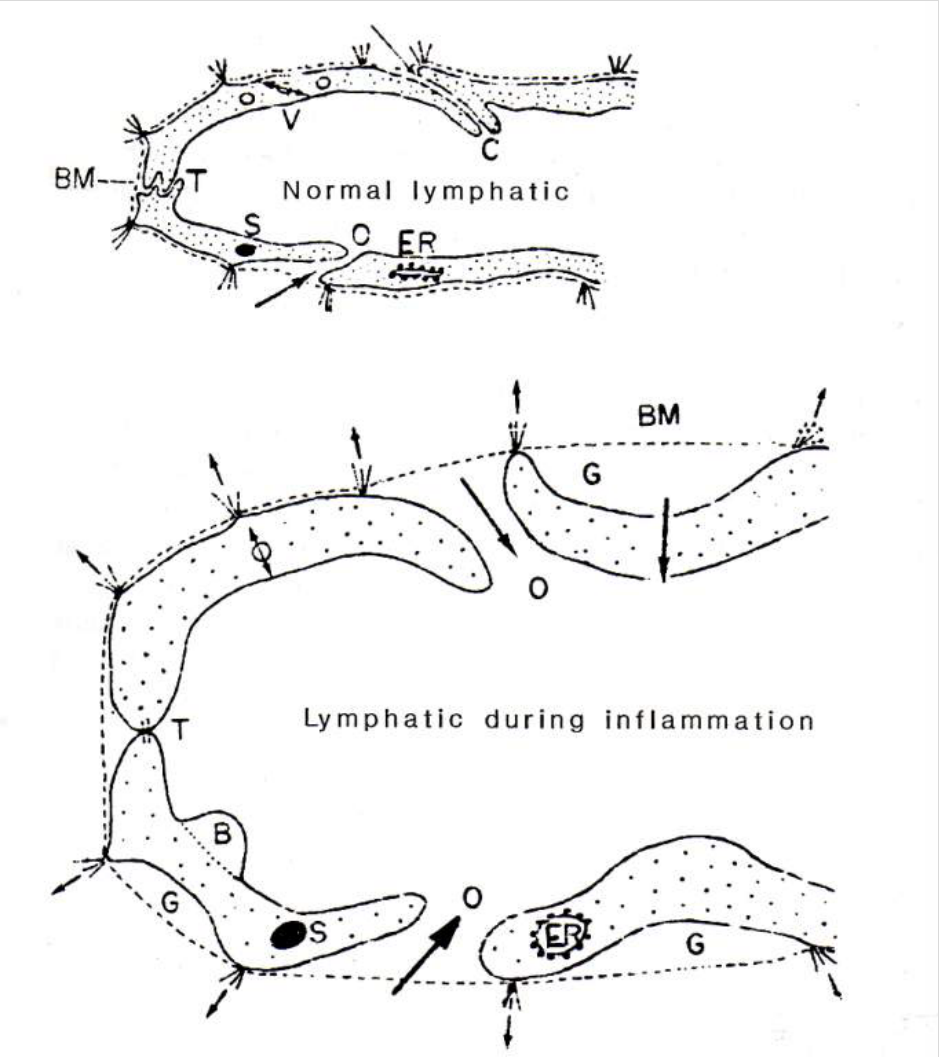
- The hydrostatic gradient that producse capillary filtration dissipates almost immediately beyond the edge of the vessel.



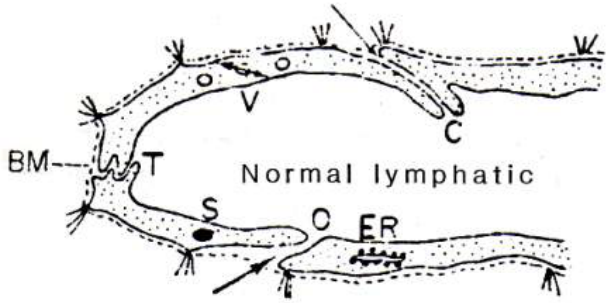
- It does not account for the movement of fluid through the interstitial matrix (gel) where cellular respiration takes place
- By the time fluid reaches the lymphatic, there is actually a small uphill gradient which must be overcome for lymph to form.

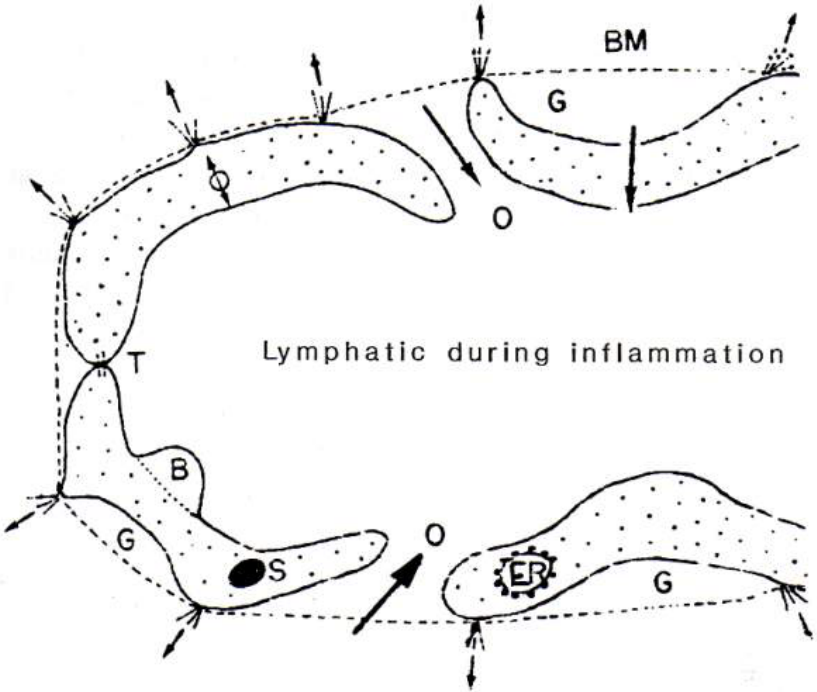
Lymph Formation

- Alternating movement
- Fluid flux



- Anchoring filaments produce a change in volume with alternating movements in the local interstitial tissue.





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Overlapping endothelial cells

Allow the ebb and flow of a fluid fluctuation to move fluid from the interstitium to the lymphatics

Treating the local response

- Improve lymph circulation
  - Respiratory/circulatory function
  - Remove obstruction to lymph flow
  - Promote lymph formation – lymph pump

Clinical applications

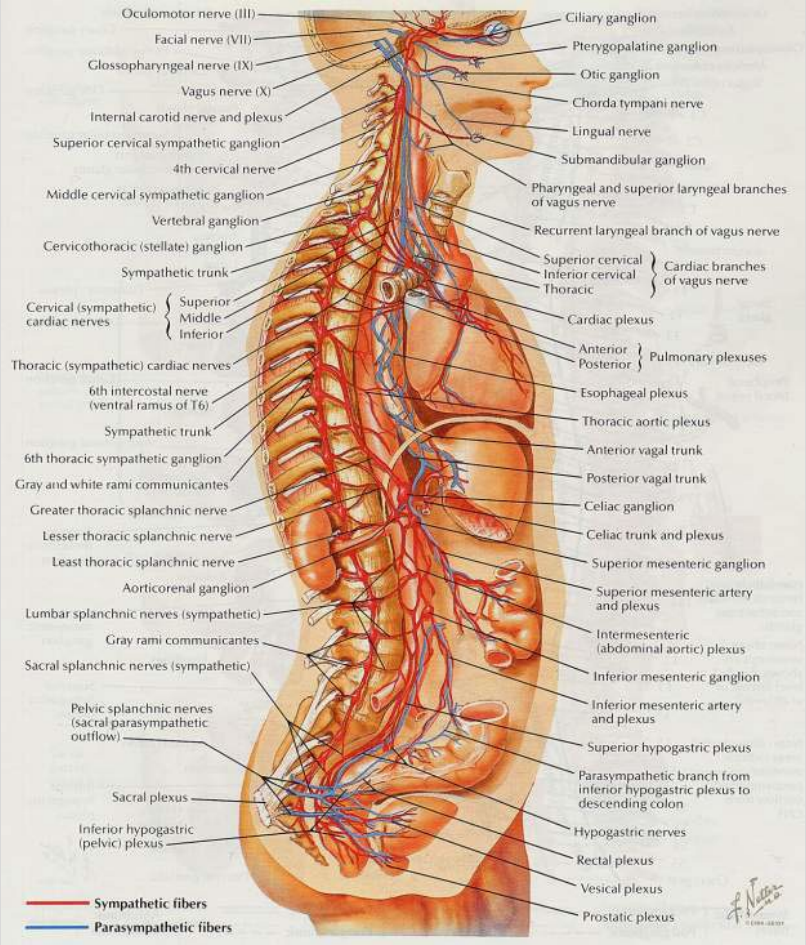
- Asthma
- All infections
- Rheumatic diseases
- All post surgical cases
- Few disease process occur without a local inflammatory process

Indications for treatment

- Clinically and/or palpably inflamed tissue.
- Palpable and/or clinical absence of intrinsic tissue and or fluid movement

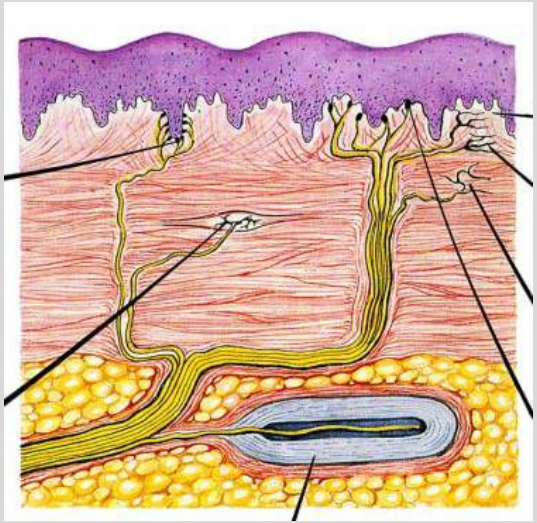
The spinal cord response

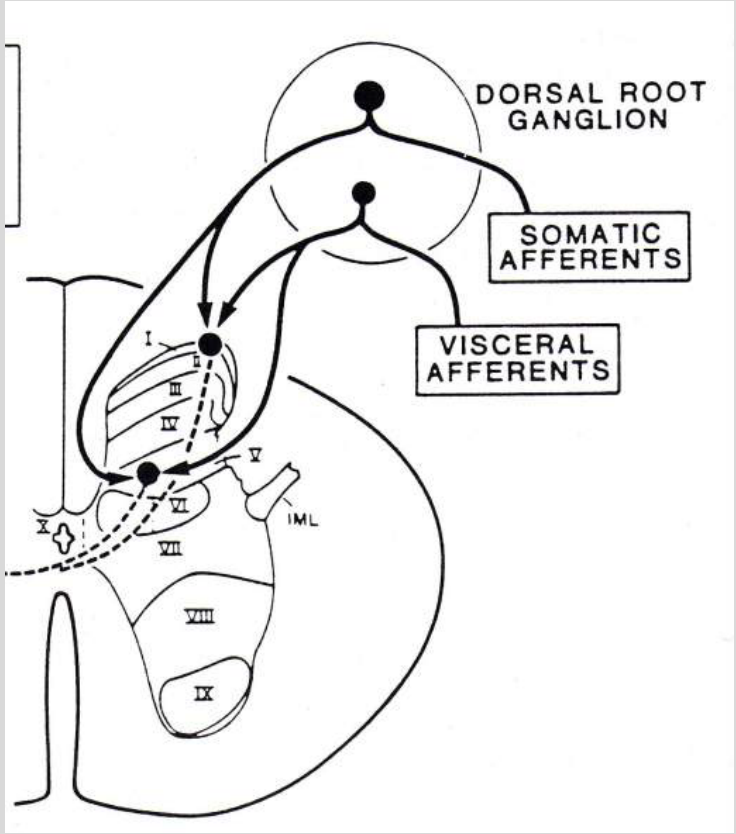
- Inflammation is a powerful stimulator of local nociceptors.
- Nociception produces a spinal reflex response, the viscerosomatic and somatovisceral reflexes.


Oculomotor nerve (III)
Facial nerve (VII)
Glossopharyngeal nerve (IX)
Vagus nerve (X)
Internal carotid nerve and plexus
Superior cervical sympathetic ganglion
4th cervical nerve
Middle cervical sympathetic ganglion
Vertebral ganglion
Cervicothoracic (stellate) ganglion
Sympathetic trunk
Cervical (sympathetic) cardiac nerves
Superior
Middle
Inferior
Thoracic (sympathetic) cardiac nerves
6th intercostal nerve (ventral ramus of T6)
Sympathetic trunk
6th thoracic sympathetic ganglion
Gray and white rami communicantes
Greater thoracic splanchnic nerve
Lesser thoracic splanchnic nerve
Least thoracic splanchnic nerve
Aorticorenal ganglion
Lumbar splanchnic nerves (sympathetic)
Gray rami communicantes
Sacral splanchnic nerves (sympathetic)
Pelvic splanchnic nerves (sacral parasympathetic outflow)
Sacral plexus
Inferior hypogastric (pelvic) plexus
Sympathetic fibers
Parasympathetic fibers
Ciliary ganglion
Pterygopalatine ganglion
Otic ganglion
Chorda tympani nerve
Lingual nerve
Submandibular ganglion
Pharyngeal and superior laryngeal branches of vagus nerve
Recurrent laryngeal branch of vagus nerve
Superior cervical
Inferior cervical
Thoracic
Cardiac branches of vagus nerve
Cardiac plexus
Anterior
Posterior
Pulmonary plexuses
Esophageal plexus
Thoracic aortic plexus
Anterior vagal trunk
Posterior vagal trunk
Celiac ganglion
Celiac trunk and plexus
Superior mesenteric ganglion
Superior mesenteric artery and plexus
Intermesenteric (abdominal aortic) plexus
Inferior mesenteric ganglion
Inferior mesenteric artery and plexus
Superior hypogastric plexus
Parasympathetic branch from inferior hypogastric plexus to descending colon
Hypogastric nerves
Rectal plexus
Vesical plexus
Prostatic plexus

Nociceptors

- Produce the afferent “drive” which initiates and maintains the reflex
- Sensitivity to inflammatory mediators, “chemonociception”, produces the exaggerated response leading to facilitation
- Locally release proinflammatory mediators which amplify the local inflammatory process





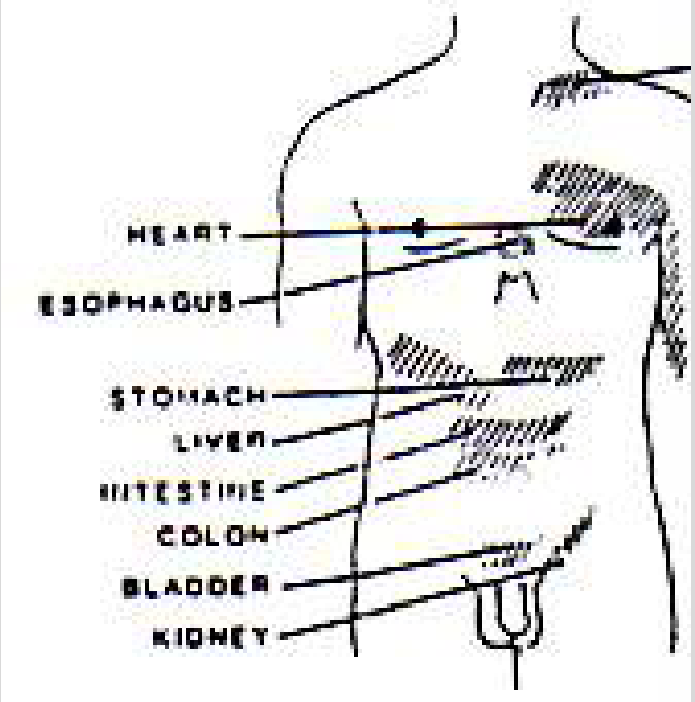
Convergence

- The convergence of visceral nociceptors with the nociceptors from all somatic tissues produces several clinical effects:
  - Referred pain
  - Segmental facilitation at the spinal cord level

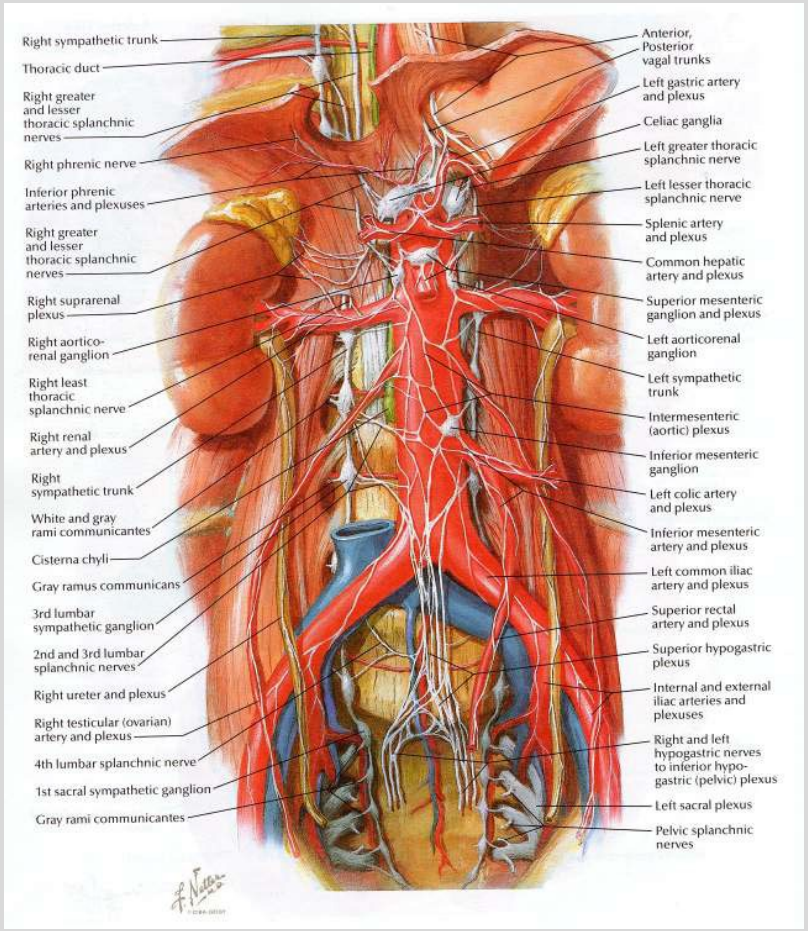
Spinal cord facilitation

- Reduced threshold for firing of the interneurons receiving nociceptive input
- Exaggerated segmental autonomic and alpha-motor response; produces boggy spasm, increased temperature, increased sweat.
- Exaggerated ascending tract input to higher centers, produces hyperasthesia, and referred pain
- Alters autonomic outflow to viscera

Clinical Applications - Diagnosis

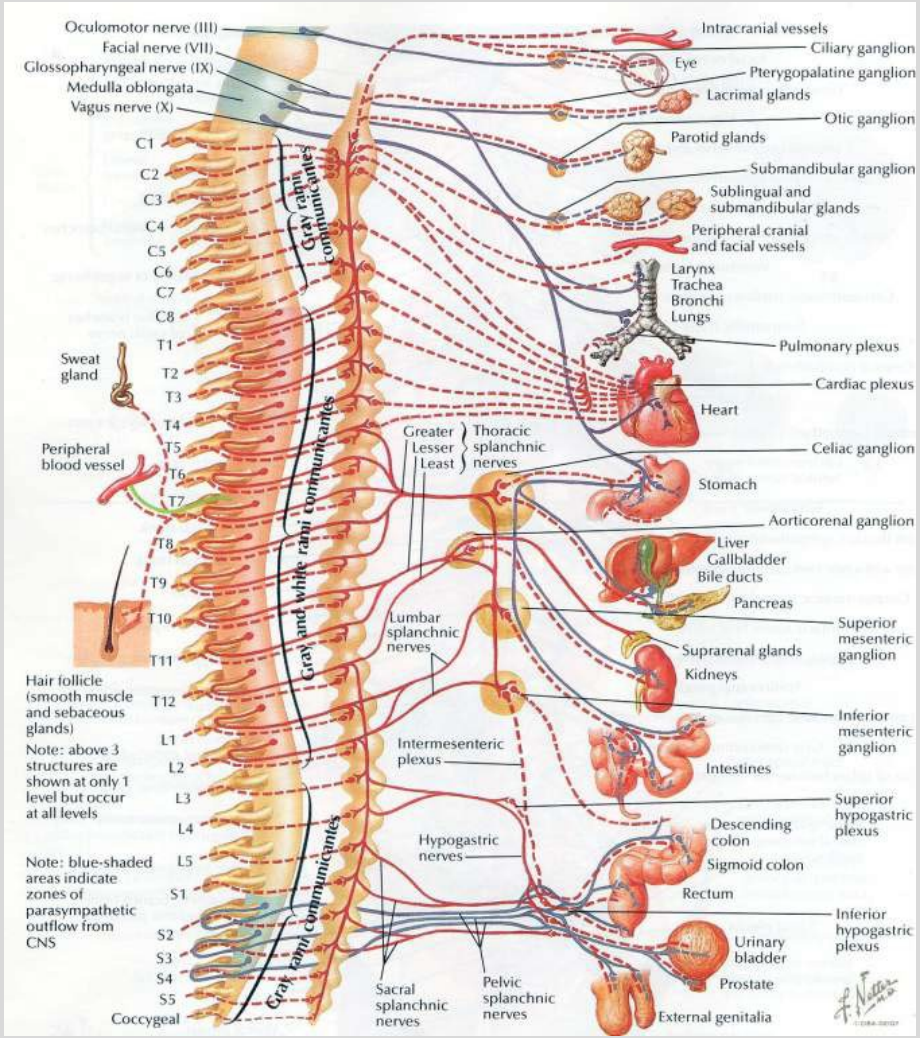


- Referred pain
- Identifying the presence of an acute process
- Identifying the diseased organ system
- Monitoring the severity of the disease
- Does this suggest limits of neural control



The innervation of all abdominal and pelvic viscera comes through 3 nerves and 3 ganglia.

1. Greater splancnic nerve >celiac ganglion, T5 – T9
2. Lesser splancnic nerve >superior mesenteric ganglion, T10 – T12
3. Least splancnic nerve>inferior mesenteric ganglion, T12 – L2



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T6-T9: R: liver, gall bladder, duodenum, pancreas,

L: spleen, stomach, pancreas

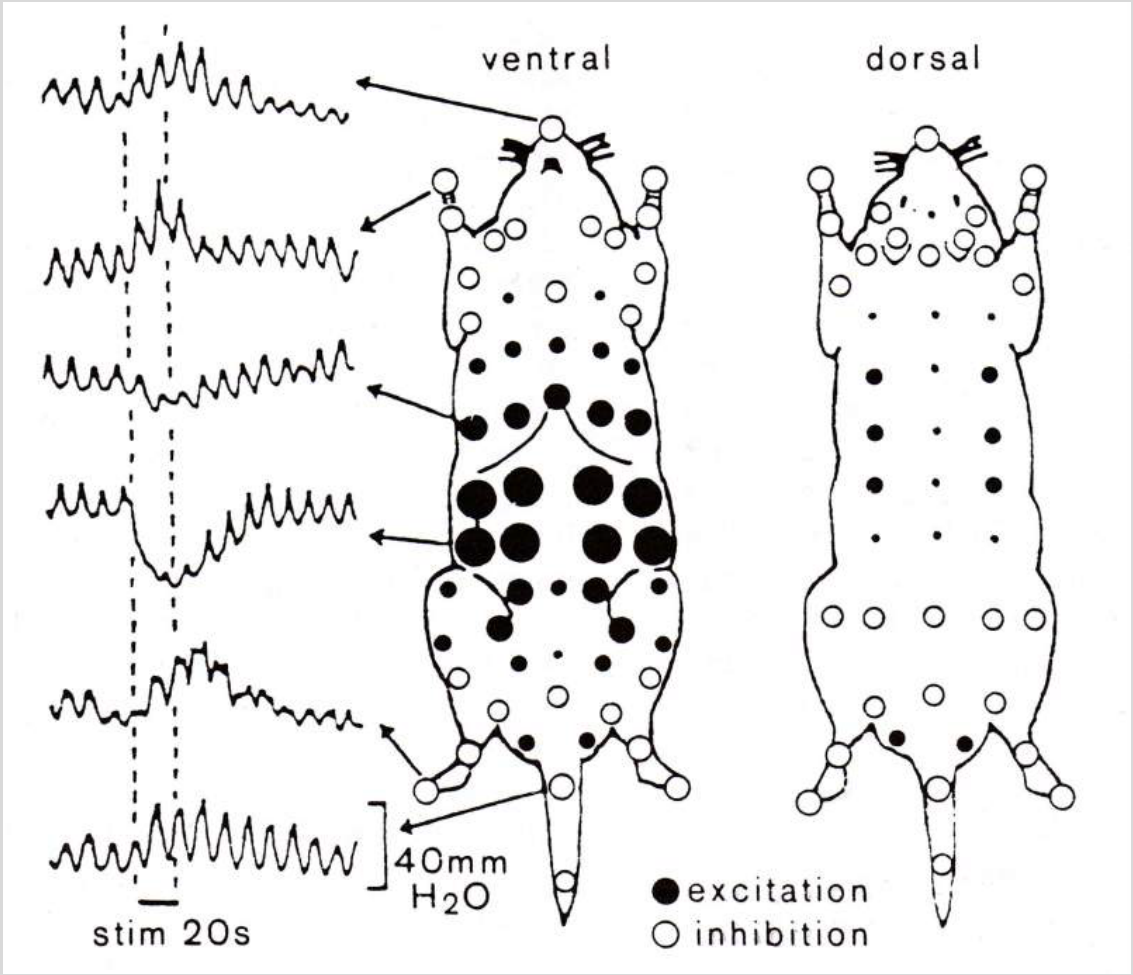
T10-T12: Sm. Bowel through transverse colon, kidney &proximal ureter, ovaries & proximal fall. Tubes, testes.

T12-L2: Bladder & distal ureter, sigmoid colon-rectum, uterus, prostate, distal fallopian tubes

Clinical applications - Treatment

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- Reducing segmental facilitation will improve neural control of the tissue
  - G.I.
    - Ileus
  - Lungs
    - Asthma
  - Cardiac
    - Ischemic heart disease/MI


ventral
dorsal
40mm
H_2O
stim 20s
excitation
inhibition

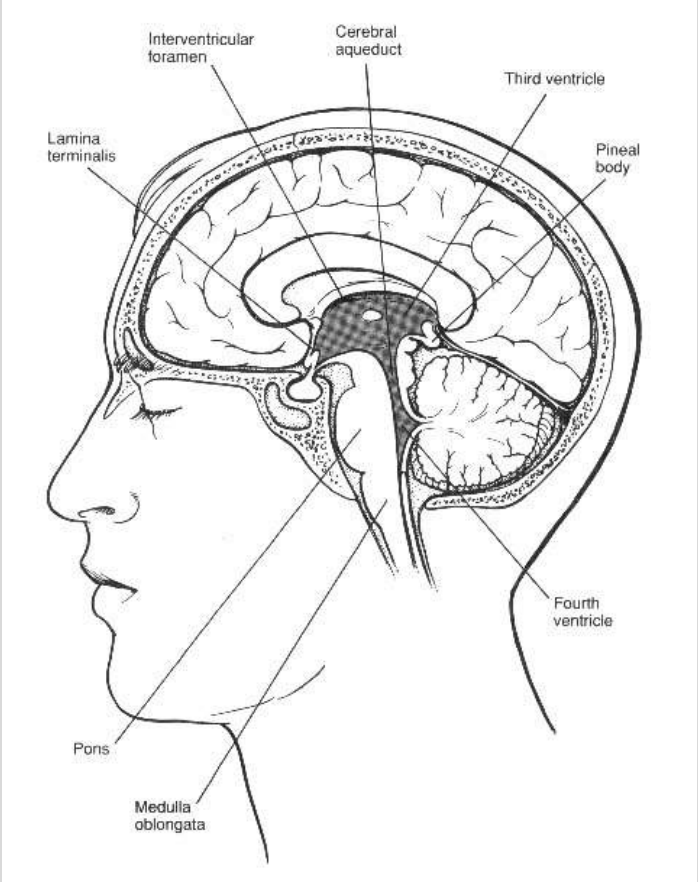
Treating segmental facilitation

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- No techniques that are painful or involve rapid movements
- Treat facilitated areas first
- Direct treatment to osseous tissues
- Anatomical location of ganglia
- May require frequent treatment with acute process

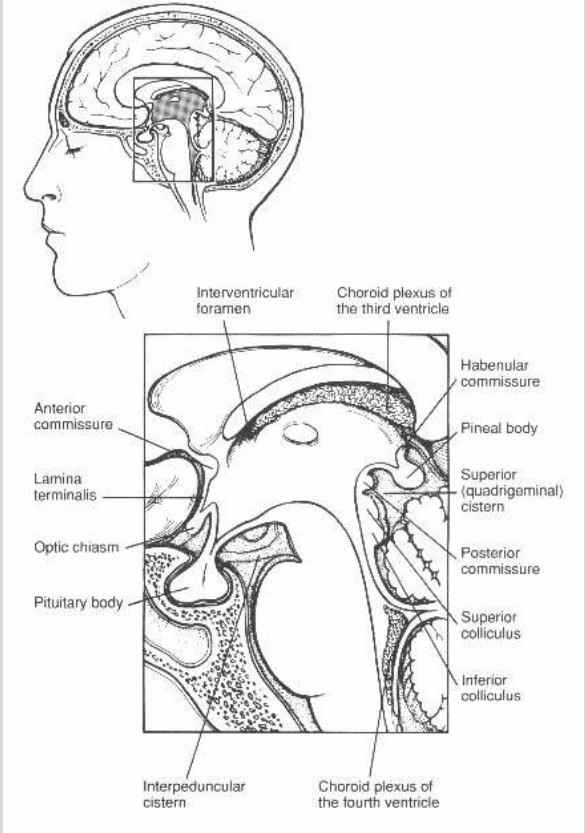
The central response

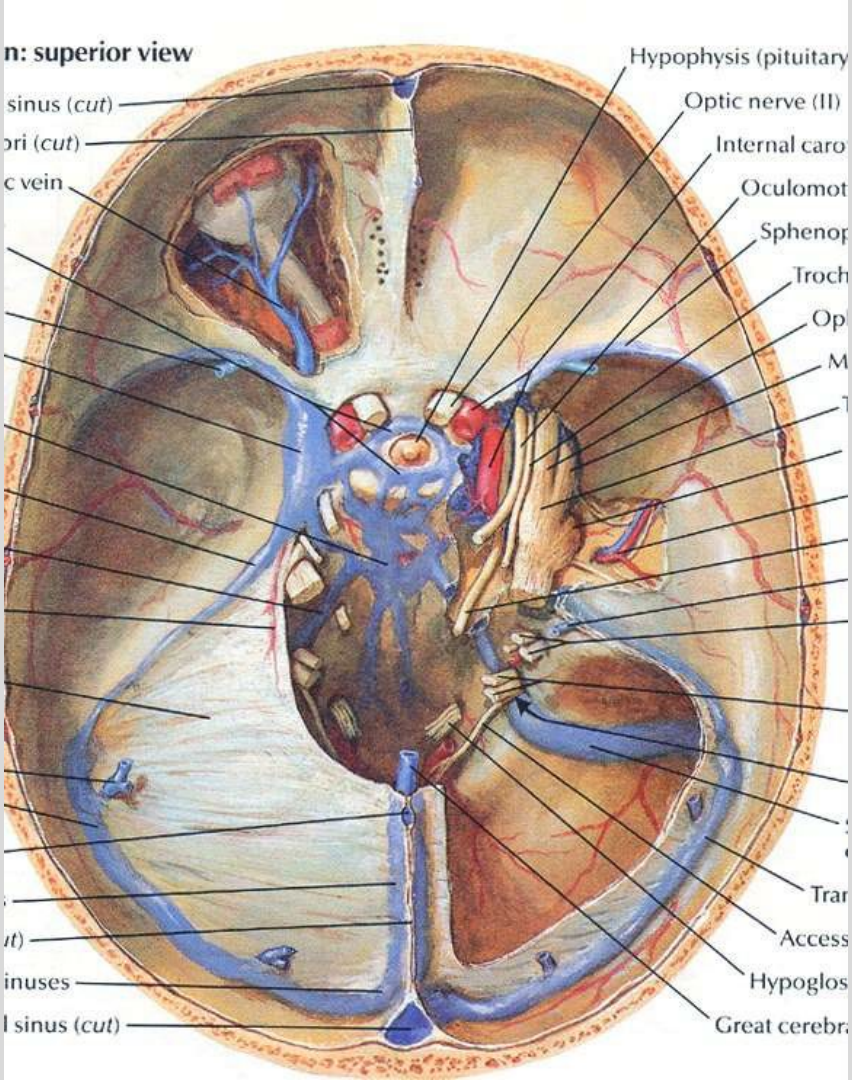
- Ascending tracts reach the hypothalamus and are a powerful stimulator of the HPA axis



The central response

- Cytokines from the local process cross the blood brain barrier and will also stimulate the HPA axis
- The central response will influence both the autonomic nervous system and the immune system
- Somatic dysfunction may influence the function of the HPA axis.





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Movement of the cranium will change the shape of the pituitary, and move blood through it's portal circulation.

Evaluation and treatment of the cranium may enhance the function of the HPA axis

Conclusions

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- The body responds to a disease process locally, at the spinal cord level, and centrally with the HPA axis
- The osteopathic structural exam may identify at which levels this process is not functioning well.
- This should help with the development of the overall treatment plan for the patient

References

- Foundations of Osteopathic Medicine 4th Edition by Seffinger. Chapter 40 A. pg 923-944

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